

UPSC Prelims

Science & Tech

Class No. 6

Health Issues

Genome Sequencing



- Recently, an international team described the first ever sequencing of complete human genome.
- Previous effort : couldn't read 8% of the genome.
- Previous effort was called Human Genome Project

Human Genome Project



- Started in 1990
- 3 billion base pairs of human DNA
- Ethical legal and social issues were addressed
- 97% Junk DNA, only 3% codes for protein
- HGP read : India did not participate
- Built a standard human resource consortium called Genome reference consortium build 38 (GRCch38)
- HGP write: started in 2016 : build an artificial human genome

Human Genome Project : uses



- Disease diagnosis : Identification of genes, rare diseases, neurodegenerative conditions.
- Viral Diseases and their impact (we now have a reference)
- Study of Genetic Variations
- Racism : Debunked
- Gene Therapy
- Evolutionary Biology : Our relationship with primates

Telomere to Telomere Project



- HGP did not do sequencing for 8% of human genome. This was heterochromatin. It was thought to be junk DNA and difficult to sequence.
- Now T2T project has done sequencing for heterochromatin region also.
- Significance :
- Genetic Variations in heterochromatin
- Understanding how genome as a whole works.
- Genetic Diseases
- Pharmacogenetics.

CHROMOSOME PARTS

HETEROCHROMATIN:

- More condensed
- Silenced genes (methylated)
- Gene poor (high AT content)
- Stains darker

EUCHROATIN

- Less condensed
- Gene expressing
- Gene rich (high GC content)
- Stains lighter



Telomere

Indian Initiatives



- CSIR : Indigen Project : Whole genome sequencing of 1000 Indian individuals representing diverse ethnic groups.
- DBT : Genome India project : 10000 genetic samples from citizens across india to build a reference genome.

Viruses



- Borderline between living and non living
- The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.
- A virus is a protein and genetic material that is infectious.
- In general, viruses that infect plants have single stranded RNA and viruses that infect animals have either single or double stranded RNA or double stranded DNA.
- Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses

Viruses



- The protein coat called capsid made of small subunits called capsomeres, protects the nucleic acid.
- These capsomeres are arranged in helical or polyhedral geometric forms.
- Viruses can cause many diseases in plants and animals.
- **Viroids** : In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight.

Viral Vector Vaccines vs DNA Vaccines



- Viral Vector Vaccine : Delivers genetic Material (DNA) that can be transcribed by the recipient's host cells as mRNA to later be translated into antigen to elicit an immune response.
- DNA Vaccine : nucleic acid vaccines that are genetically engineered into a plasmid. The DNA sequence codes for the antigen against which immunity is sought.

DNA Vaccine vs Recombinant Vaccine



DNA Vaccine	Recombinant Vaccine
Description	
DNA vaccines are prepared by inserting a DNA sequence (that codes for the antigen) into a plasmid and transfected to the host cell.	Recombinant vaccines are protein or DNA recombinants that are inserted into the host cell to trigger immune response.
DNA/Protein	
DNA-based.	Both DNA and protein-based.
Incorporation	
The DNA sequence gets incorporated into the host genome.	A gene of interest is incorporated into a vector which in turn is included in the host genome.
Approval	
ZyCoV-D is a DNA vaccine approved for humans.	Hepatitis B and HPV vaccines have been approved and are readily used.

Nutrients and their roles



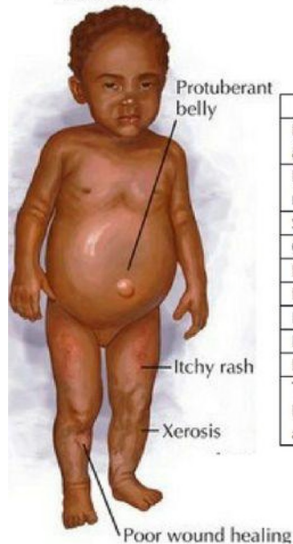
- Major nutrients : Carbohydrates , Proteins, Fats
- Doubt regarding proteins
- Major deficiencies and issues
- Kwashiorkor and Marasmus

Kwashiorkor and Marasmus



10 Differences between Kwashiorkor and Marasmus

Kwashiorkor

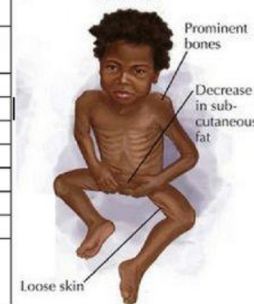


Comparison Table

Kwashiorkor	Marasmus
It develops in children whose diets are deficient of protein.	It is due to deficiency of proteins and calories.
It occurs in children between 6 months and 3 years of age.	It is common in infants under 1 year of age.
Subcutaneous fat is preserved.	Subcutaneous fat is not preserved.
Oedema is present.	Oedema is absent
Enlarged fatty liver.	No fatty liver.
Ribs are not very prominent.	Ribs become very prominent.
Lethargic	Alert and irritable.
Muscle wasting mild or absent.	Severe muscle wasting
Poor appetite.	Voracious feeder.
The person suffering from Kwashiorkor needs adequate amounts of proteins.	The person suffering from Marasmus needs adequate amount of protein, fats and carbohydrates.



Marasmus



Kwashiorkor vs Marasmus

Micronutrients and their deficiencies



- Certain other nutrients are also important for the body
- Called micro-nutrients because they are needed in much more quantities than macro-nutrients
- Include vitamins and minerals
- We need to remember certain names of micronutrients and diseases caused by them : no option but to mug this up

Micronutrient Diseases



Vitamin	Chemical Name	Deficiency Diseases
Fat soluble Vitamins		
A	Retinol, Retinal, Retinoic acid	Night-blindness and keratomalacia
D	Ergocalciferol (D ₂), Cholecalciferol (D ₃)	Rickets and Osteomalacia
E	Tocopherol	Mild hemolytic anemia in newborn infants
K	Phylloquinone (K ₁), Menaquinones (K ₂)	Bleeding diathesis
Water soluble vitamins		
B ₁	Thiamine	Beriberi
B ₂	Riboflavin	Ariboflavinosis
B ₃	Niacin, Niacinamide	Pellagra
B ₅	Pantothenic acid	Paresthesia
B ₆	Pyridoxine, Pyridoxamine, Pyridoxal	Anemia peripheral neuropathy
B ₇	Biotin	Dermatitis
B ₉	Folic acid, Folinic acid	Neural tube defects
B ₁₂	Cyanacobalamine	Megaloblastic anemia
C	Ascorbic acid	Scurvy

Diseases Caused by various Pathogens



- Question : Differentiate between Patho-gen and Anti-gen
Hint : word association will help
- Diseases :
- Viral
- Bacterial
- Protozoan
- Fungi
- Role of Vectors : Question : what is the meaning of vector?

Types of Transmissions



- Feco – oral
- Droplet
- Blood / serum / sexual
- Zoonotic
- Wound breaks

Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Influenza	RNA virus	Respiratory passages: epithelial lining of trachea and bronchi.	Droplet Infection	AVAILABLE : constantly developed due to mutations
Common cold	rhino-virus (RNA Virus)	Respiratory Passage	Think? –do-	AVAILABLE : Flu shots
Small pox	RNA virus	Skin (visible) and Respiratory pathway	-do-	First ever vaccine : Edward Jenner, Live attenuated, now not needed.
Chickenpox	Virus	Blistering Skin rash	Air-borne droplets	Living attenuated virus
Mumps	Virus (RNA virus)	Respiratory Passage	Droplet	LAV
Measles	RNA virus	Lungs, then spread to skin	-do-	-do-
Rubella (german measles)	Virus	Respiratory passage and lymph nodes	Droplet	MMR vaccine for all 3

Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Polio	Rna virus	Nervous system : leads to paralysis	Feces and droplets : Role of contamination of water and hygiene	Oral Polio Vaccine (OPV) IPV : injectable
Yellow Fever	RNA virus	Mild : fever, later fatal heart liver and kidney conditions	Vector based spread : aedes mosquito	LAV
AIDS	RetroVirus (RNA)	Human Immunodeficiency : destroys immune system	Sexual Intercourse, Blood, Vertical Transmission	NONE
Ebola	Ebola Virus (RNA)	Chills, Dehydration, fever. Ultimately : Fatal	Zoonotic Disease and later through contact, even with dead	NONE
Zika	Virus (RNA)	Mild illness, dengue like symptoms	Vector Bite : aedes, sometimes sexual	NONE
Rotavirus	RNA	Diarrhea : GI tract	Think?	Yes, recently developed
Hepatitis : Important, next slide	Word Breakdown			

DIFFERENCE BETWEEN IPV AND OPV

IPV	OPV
Killed formolised virus	Live attenuated virus
Given IM/SC	Given orally
Induces circulating antibody; no local immunity	Both humoral and intestinal immunity
Prevents paralysis; does not prevent reinfection by wild polio viruses	Prevents paralysis and intestinal reinfection
Not useful in epidemics	Effective in controlling epidemics
Content is 10,000 times more than OPV; Costlier	Cheaper
Does not require stringent conditions during storage and transportation	Requires to be stored and transported at sub-zero temperature, unless stabilised

Type of Hepatitis

	A	B	C	D	E
Source of virus	feces	blood/ blood-derived body fluids	blood/ blood-derived body fluids	blood/ blood-derived body fluids	feces
Route of transmission	fecal-oral	percutaneous permucosal	percutaneous permucosal	percutaneous permucosal	fecal-oral
Chronic infection	no	yes	yes	yes	no
Prevention	pre/post- exposure immunization	pre/post- exposure immunization	blood donor screening; risk behavior modification	pre/post- exposure immunization; risk behavior modification	ensure safe drinking water

Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Japanese Encephalitis(JE)	ssRNA	Brain swelling : severe headache. Fatal in children	Culex	Available, Part of National Schedule
KFD	ssRNA	High grade fever and chills	Tick, but no human to human	yes

Viral Diseases in News



- **Zombie Virus** : Zombie virus is virus emerged due to thawing of permafrost as global temperature is rising. The oldest virus is 48,500 years old, called Pandoravirus yedoma. Pandoravirus was discovered below the bottom of a lake in Yukechi Alas in Yakutia, Russia.
- **Marburg Virus Diseases** : Ghana first case. It's a highly infectious viral haemorrhagic fever with a fatality ratio of up to 88 percent. Fruit bats of Pteropodidae Family are key carriers of the disease. Human to h
- **African Swine Flu** : viral disease affecting pigs, not humans.
- **Tomato Flu**
- **Lumpy Skin Disease** : Lumpiprovac-ind developed by ICAR, Live Attenuated Vaccine, Vector Borne Diseases in Cattle, Not Zoonotic

Viral Diseases in News



- White Spot Syndrome Virus : DNA virus, high level of mortality in shrimp
- Porcine virus : Death due to Xenotransplantation (pigs)
- Canine Distemper Virus : Vaccine developed for Asiatic lion.
- Acute Encephalitis Syndrome : Fever and change of Mental State

Covid 19 Mutation



- Covid Virus has been mutating and new variants are forming.
- We know what mutations are : lead to changes in gene of virus.
- New Variant : Functional properties different from original virus.
- When new variant gets established in population, it is referred to new strain. A strain behaves differently from parent virus.
- All strains are variants but not all variants are strains.

Covid 19 Mutation



- Variant classifications :
- Variant of Concern : Increased transmissibility, increased fatality, significant decrease in vaccine or therapy effectiveness. Alpha, Beta, Gamma, Delta, Omicron
- Variant of Interest : Disease severity, immune escape, transmissibility, diagnostic escape, and increased volume of community transmission. Eta, Kappa, Lambda
- Variant Under Monitoring : Genetic changes are there but epidemiological impact is unclear.

Covid 19 Mutation



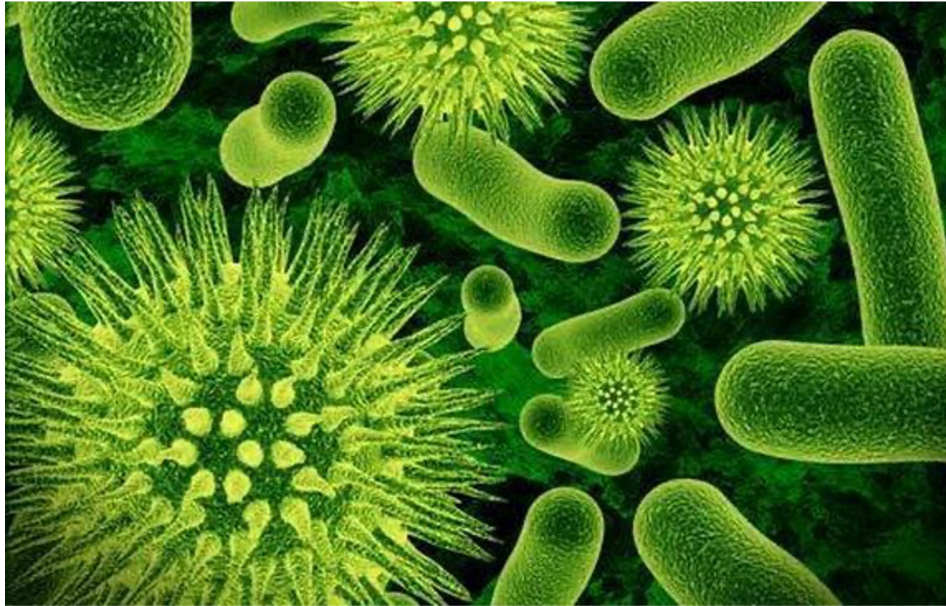
- 2 important things : Receptor that covid binds to : ACE 2 : Angiotensin Converting Enzyme 2 : Mutation here leads to change in covid infection ability
- Place where antibody binds to covid : Mutation leads to antibody ability
- Site 486 : Controls both things at once
- XBB, XBB.1 : Change in nucleotide : Less bind, More escape.
- XBB 1.5 : 2 mutations simultaneously, Decrease in bind is much less, but antibody escape is much higher. Very dangerous: in china
- Related term : Hybrid Immunity (i'll explain)

Influenza Virus



- Cause respiratory infection.
- 4 types of seasonal viruses : A,B,C,D. A and B cause seasonal epidemics.
- Influenza A : classified on basis of proteins on surface of virus. Hemagglutinin and Neuraminidase. H1N1 and H3N2 are currently circulating. These are swine influenza.
- H9N2, H5N1 are avian influenza.

Bacterial Diseases



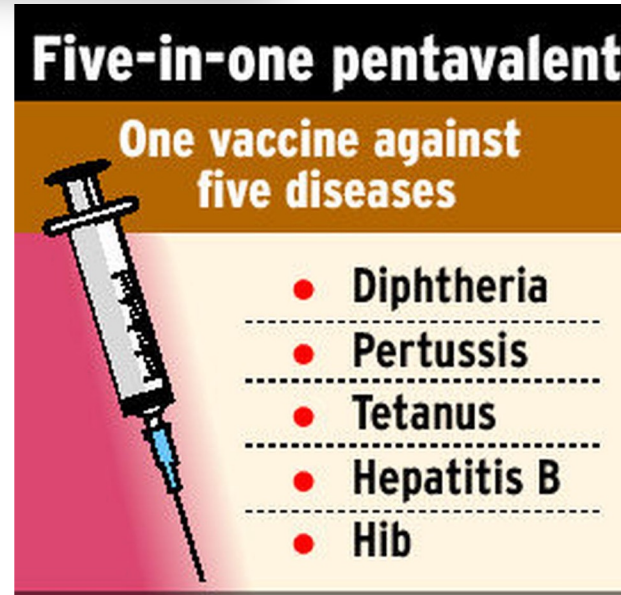
Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Diphtheria	Corynebacterium diphtheria	Respiratory Tract	Use logic	Toxoid vaccine
Tuberculosis (TB)	Mycobacterium tuberculosis	Mainly lungs	Droplet infection, Drinking milk from infected cattle.	BCG living attenuated bacteria. Antibiotics e.g. streptomycin.
Pertusis (whooping cough)	Bordetella Pertusis	Respiratory Tract	Droplet	Killed Vaccine
Gonorrhea	N. Gonorrhea	Reproductive Tract, blindness in infants	STD, vertical	No vaccine, but treatable through AMR
Syphilis	T. Pallidum	Reproductive Tract, later other organs including brain	STD	-do-
Tetanus	Clostridium Tetani	Blood : Toxin : later lock jaw	Wound Breaks	Toxoid : done already
Cholera	Vibrio Cholera	GI tract : diarrhoea,	Oro-faecal : water / objects / vector	Antibiotics

Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Typhoid	Salmonella Typhi	GI tract then other organs	Oro Faecal : water / objects / vector	Ty21a and Vi capsular polysaccharide vaccine
Bacterial Dysentery	Shigella	GI tract	-do-	None, but treatable
Haemophilus Influenza type b (HiB) disease	H. Influenzae	Lungs : pneumonia	Respiratory droplets	YES, in national schedule

Pentavalent Vaccine



- Again name breakdown : Penta means 5



PROTOZOAN DISEASES



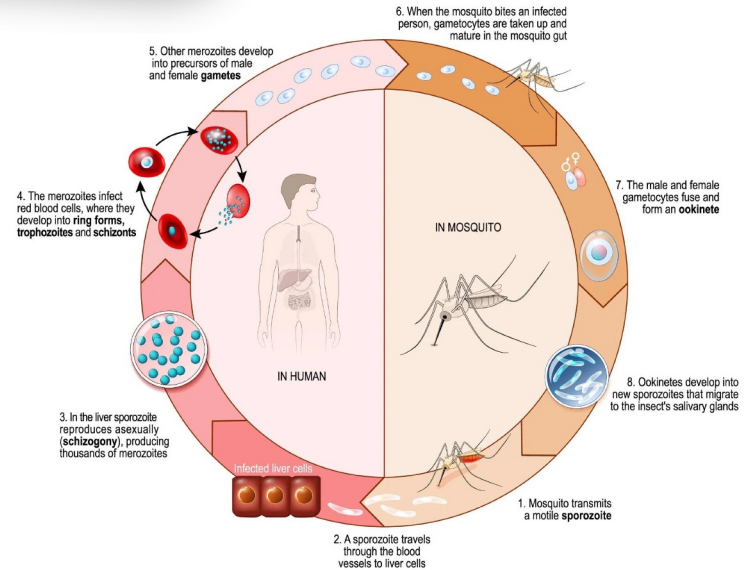
- Filariasis : caused by roundworms
- Vector : Culex, Aedes and Anopheles mosquitoes
- Swelling of lymph nodes : also called elephantiasis
- Lymph?



Malaria



- Caused by plasmodium
- Vector : anopheles
- 4 types : vivax / malaria / falciparum / ovale
- Most severe : falciparum, vivax
- Both found in India



Malaria



- World Malaria Report : Released by WHO
- Malaria Vaccine
- WHO has recommended broad use of the RTS,S/AS01 (RTS,S) malaria vaccine among children living in regions with moderate to high *P. falciparum* malaria transmission.
- Known by its brand name **Mosquirix**, this recombinant protein vaccine has been shown to significantly reduce malaria, and deadly severe malaria among young children, and also reduced the overall hospital admissions.
- India related findings
 - o India accounted for 83% of cases in the WHO South-East Asia Region.

Initiatives for Malaria



- **Malaria Elimination Research Alliance (MERA)- India:** Its purpose is to identify, articulate, prioritise and respond to the research needs of the country to eliminate malaria from India by 2030.
- **E-2025 initiative:** Under this, WHO has identified group of 25 countries with the potential to eradicate malaria by 2025.
- **High Burden to High Impact (HBHI) initiative:** In 2019,WHO has initiated the HBHI initiative in 11 high malaria burden countries, including **India** (West Bengal and Jharkhand, Chhattisgarh and Madhya Pradesh).
- **National Strategic Plan for Malaria Elimination (2017-22)**
- **Ministerial Declaration on Accelerating and Sustaining Malaria Elimination in South-East Asia Region (SEAR)**

Kala Azar



- Aka Leishmaniasis
- Vector : Sandfly
- Endemic to Indian Subcontinent
- Hepato-Splenomegaly is the most common feature

Fungi



- **Diseases caused by Fungi-**
- Ringworm,
- Athlete's foot
- Dhobie itch
- Madura foot

Common Vectors



- **Aedes** – Chikungunya, Dengue fever, Lymphatic filariasis, Rift Valley fever, Yellow fever, Zika
- **Anopheles** – Malaria, Lymphatic filariasis.
- **Culex** – Japanese encephalitis, Lymphatic filariasis, West Nile fever

Remember

- Sandfly?
- Tick?

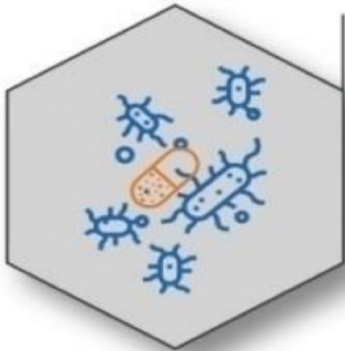
Antimicrobial Resistance



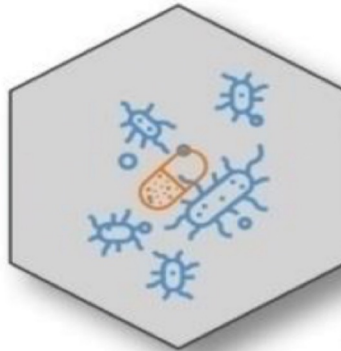
- According to a study by Lancet, Antimicrobial resistance (AMR) is now the leading cause of death across the world.
- **AMR** is the ability of a microorganism (like bacteria, viruses, and some parasites) to stop an antimicrobial (such as antibiotics, antivirals and antimalarials) from working against it.
- While resistance levels continue to grow, **antibiotic development has not.**
 - While 63 new antibiotics were approved between 1980 and 2000, **only 15 new drugs have been approved between 2000 and 2018.**

Out of the seven deadliest drug-resistant bacteria, **vaccines are only available against bacteria causing pneumonia and tuberculosis.**

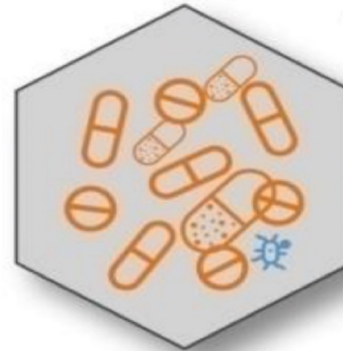
HOW ANTIBIOTIC RESISTANCE HAPPENS



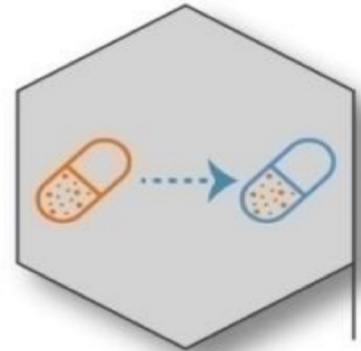
Our body is **home to countless microbes**. Some may be resistant to antibiotics



Antibiotics kill the bacteria
Causing the infections as well as the good bacteria



The **antibiotic-resistant bacteria** are now able to **grow and take over**

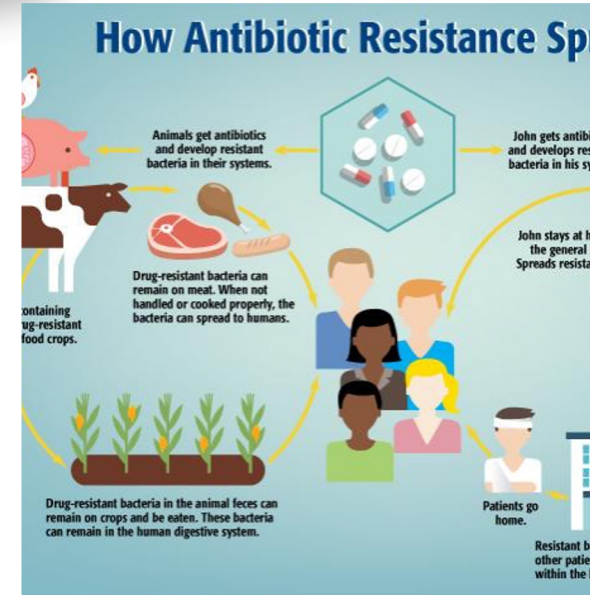


Some bacteria may **give** their antibiotic **resistance to other bacteria**

Causes of AMR



- **Poor hygiene** and infection prevention and control.
- **Overuse of antimicrobials**
- **Over prescription** by doctors and **Non- prescription purchase** in many countries.
- If a person **does not complete a course of antimicrobial drugs**, some microbes may survive and develop resistance to the drug.
- Practice of **adding antibiotics to agricultural feed** promotes drug resistance.
- **Environment** plays a significant role wherein **waste from farms, factories, community (such as mass bathing) and healthcare settings** can contribute to AMR through environmental routes.



Indian Initiatives against AMR



- **National programme on AMR containment**
- **National Action Plan on Antimicrobial Resistance (NAP-AMR)**
- **Red Line campaign**
- **Integrated One Health Surveillance Network for Antimicrobial Resistance**
- **FSSAI has set certain guidelines limiting the antibiotics in food products**
- **India's medical societies adopted the Chennai Declaration**
- **separate Schedule H-1**
- **Use bacteriophages, or 'phages', to kill bacterial cells.**

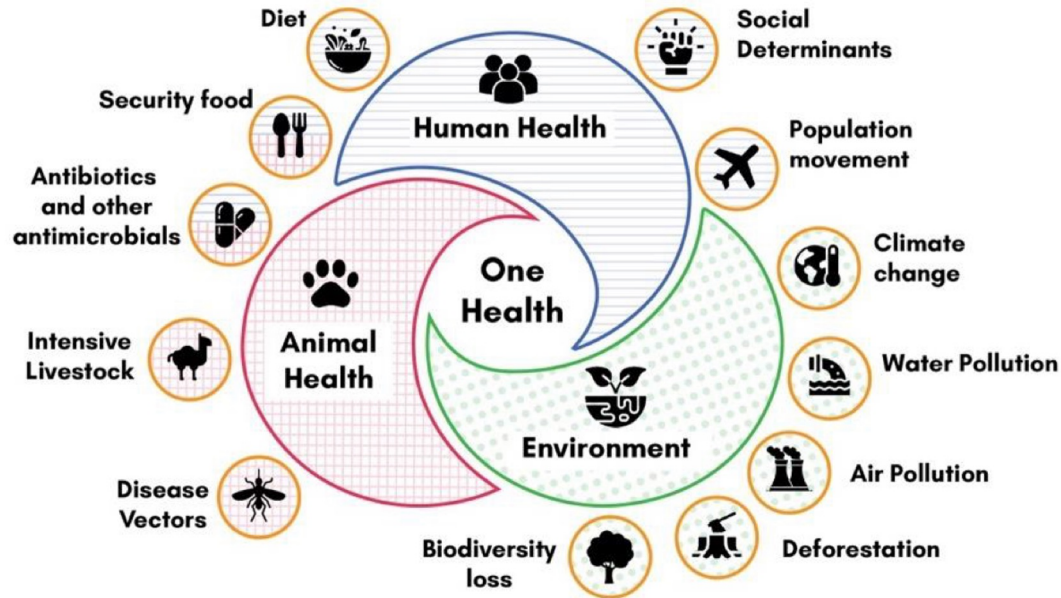
One health approach



- **FAO, the World Organisation for Animal Health (OIE), WHO and UNEP (quadrupartite)**
- These 4 organizations are **working together to mainstream One Health, they have recently launched the One Health Joint Plan of Action (2022-2026)**
- One Health is an **integrated, unifying approach** that aims to sustainably balance and **optimise the health of people, animals and ecosystems.**
- **Benefits of One Health:**
- **Promote coherence** in action across sectors and areas of expertise in health sector.
- Address the full spectrum of disease control - **from disease prevention to detection, preparedness, response, and management.**

One Health

Human health and animal health are interdependent.
At the same time, both depend on the environment.



One Health Approach : India's initiatives



- One of the priorities of our G20 Presidency
- One Health Support Unit (OHSU) by Department of Animal Husbandry and Dairying, to develop a national One Health Framework.
- Resource allocation and policy ecosystem of zoonotic diseases.
- National Mission on Biodiversity and Human Well-being: Launched by Prime Minister's Science, Technology and Innovation Advisory Council (PM-STIAC).
- One Health' project: By Department of Biotechnology to carry out surveillance of zoonotic as well as transboundary pathogens in India.

One Health Approach : Other global initiatives



- Pilanesberg Resolution : governments and donors
- One world One Health : wildlife conservation society with 12 Manhattan principles.
- National Framework for one health : by FAO

Traditional Medicine



- WHO Director-General laid the foundation stone for world's first and only Global Centre for Traditional Medicine (GCTM) at Jamnagar in Gujarat.
- Traditional medicine (TM) is the sum of the knowledge, skill, and practices based on the theories, beliefs, and experiences indigenous to different cultures, for treatment of physical and mental illness.
- Complementary medicine (CM) or alternative medicine refers to healthcare practices that are not part of that country's own tradition or conventional medicine and are not fully integrated into the dominant health-care system.

Traditional Medicine : Indian Initiatives



- Indian Initiatives :
- National AYUSH Mission (NAM)
- Champion Services Sector Scheme for Medical value Travel
- Ayush Export Promotion Council (AEPC)
- The AYUR SWASTHYA YOJANA
- AYUSH Information Cell

Regulations for Ayurveda Aahara



- FSSAI and MoAYUSH
- Marketing of quality food products to help in expanding the international market for for Make in India
- strict Food Safety and Standards (Ayurveda Aahara) Regulations, 2022 rules.
- special logo has been created for “Ayurveda Aahara”
- ‘Ayurveda Aahara’ will not include Ayurvedic drugs or proprietary Ayurvedic medicines and medicinal products, cosmetics, narcotic or psychotropic substances and herbs.

Viral Diseases in News



- **Monkeypox** : PHEIC, renamed as mpox, zoonotic disease, belongs to orthopoxvirus which also includes variola virus (smallpox), cowpox.
- Animal to human transmission : direct contact with blood, body fluids, lesions.
- Human to Human : Sexual, blood borne.
- HPV : Cervavac', the country's first quadrivalent human papillomavirus vaccine (qHPV) approved for market authorization. **Manufactured by**: Serum Institute of India.
- HPV is a sexually transmitted virus that may cause cervical cancer. It can be eliminated.

Neglected Tropical Diseases



- Communicable diseases in tropical and subtropical countries.
- Populations living in poverty, without adequate sanitation, close contact with infectious vectors and domestic animals.
- India : world's largest burden.
- India has eliminated several NTDs including Guinea worm, Trachoma, Yaws.
- Eradication : only smallpox.

Elimination and Eradication and Control



- Disease control : No longer a public health problem.
- Elimination : Temporary phenomenon, interruption of transmission. Newer cases have stopped but continuous efforts are needed.
- Eradication : permanent, complete eradication of the agent.

Test of 13 Questions!

Test.

Cut off : 10/13



Question 1 (Previous Year Question 2016)

Q. 'Mission Indradhanush' launched by the Government of India pertains to:

- a) Immunization of children and pregnant women
- b) Construction of smart cities across the country
- c) India's own search for Earth-like planets in outer space
- d) New Educational policy

Question 2 (Previous Year Question 2017)

Q. Consider the following statements:

1. In tropical regions, Zika virus disease is transmitted by the same mosquito that transmits dengue.
2. Sexual transmission of Zika virus disease is possible.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Question 3 (Previous Year Question 2019)

Q. Which of the following are the reasons for the occurrence of multi-drug resistance in microbial pathogens in India?

1. Genetic predisposition of some people
2. Taking incorrect doses of antibiotics to cure diseases
3. Using antibiotics in livestock farming
4. Multiple chronic diseases in some people

Select the correct answer using the code given below:

- a) 1 and 2
- b) 2 and 3 only
- c) 1, 3 and 4
- d) 2, 3 and 4

Question 4 (Previous Year Question 2019)

Q.What is Cas9 protein that is often mentioned in the news?

- a) A molecular scissors used in targeted gene editing
- b) A biosensor used in the accurate detection of pathogens in patients
- c) A gene that makes plants pest-resistant
- d) A herbicidal substance synthesized in genetically modified crops

Question 5 (Previous Year Question 2019)

Q. Which one of the following statements is not correct?

- a) Hepatitis B virus is transmitted much like HIV.
- b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.
- c) Globally, the number of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.
- d) Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years.

Question 6 (Previous Year Question 2020)

Q. Consider the following statements:

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.
2. A person's genome can be edited before birth at the early embryonic stage.
3. Human /induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 2 only
- d) 1, 2 and 3

Question 7 (Previous Year Question 2021)

Q. The term 'ACE2' is talked about in the context of

- a) genes introduced in the genetically modified plants
- b) development of India's own satellite navigation system
- c) radio collars for wildlife tracking
- d) spread of viral diseases

Question 8 (Previous Year Question 2022)

Q. Consider the following statements:

1. Biofilms can form on medical implants within human tissues.
2. Biofilms can form on food and food processing surfaces.
3. Biofilms can exhibit antibiotic resistance.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Question 9 (Previous Year Question 2022)

Q. Consider the following statements in respect of probiotics :

1. Probiotics are made of both bacteria and yeast.
2. The organisms in probiotics are found in foods we ingest but they do not naturally occur in our gut.
3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) 1 and 3
- (d) 2 and 3

Question 10 (Previous Year Question 2013)

Q. Which of the following diseases can be transmitted from one person to another through tattooing?

1. Chikungunya
2. Hepatitis B
3. HIV-AIDS

Select the correct answer using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3

Question 11 (Previous Year Question 2014)

Q. Consider the following diseases

1. Diphtheria
2. Chickenpox
3. Smallpox

Which of the above diseases has/have been eradicated in India?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3
- d) None

Question 12 (Previous Year Question 2014)

Q. Consider the following pairs :

Vitamin Deficiency disease

1. Vitamin C : Scurvy
2. Vitamin D : Rickets
3. Vitamin E : Night blindness

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3
- d) None

Question 13 (Previous Year Question 2015)

Q. H1N1 virus is sometimes mentioned in the news with reference to which one of the following diseases?

- a) AIDS
- b) Bird flu
- c) Dengue
- d) Swine flu

Congratulations on giving the test!

Let's Discuss The Answers.



Question 1 (Previous Year Question 2016)

Q. 'Mission Indradhanush' launched by the Government of India pertains to:

- a) **Immunization of children and pregnant women**
- b) Construction of smart cities across the country
- c) India's own search for Earth-like planets in outer space
- d) New Educational policy

Answer: A

- Mission Indradhanush is a health mission of the government of India. It was launched by Union Health Minister J. P. Nadda on 25 December 2014.
- It aims to immunize all children under the age of 2 years, as well as all pregnant women, against seven vaccine-preventable diseases.
- The diseases being targeted are diphtheria, whooping cough, tetanus, poliomyelitis, tuberculosis, measles and Hepatitis B.
- In addition to these, vaccines for Japanese encephalitis and Haemophilus influenzae type B are also being provided in selected states.

Question 2 (Previous Year Question 2017)

Q. Consider the following statements:

1. In tropical regions, Zika virus disease is transmitted by the same mosquito that transmits dengue.
2. Sexual transmission of Zika virus disease is possible.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2

Answer: C

- Zika virus (ZIKV) is a member of the virus family Flaviviridae.
- It is spread by daytime-active Aedes mosquitoes, such as A. aegypti and A. albopictus.
- Its name comes from the Zika Forest of Uganda, where the virus was first isolated in 1947.
- Zika virus is related to the dengue, yellow fever, Japanese encephalitis, and West Nile viruses.
- Zika can be transmitted from men and women to their sexual partners

Question 3 (Previous Year Question 2019)

Q. Which of the following are the reasons for the occurrence of multi-drug resistance in microbial pathogens in India?

1. Genetic predisposition of some people
2. Taking incorrect doses of antibiotics to cure diseases
3. Using antibiotics in livestock farming
4. Multiple chronic diseases in some people

Select the correct answer using the code given below:

- a) 1 and 2
- b) 2 and 3 only**
- c) 1, 3 and 4
- d) 2, 3 and 4

Answer: C

Question 4 (Previous Year Question 2019)

Q.What is Cas9 protein that is often mentioned in the news?

- a) **A molecular scissors used in targeted gene editing**
- b) A biosensor used in the accurate detection of pathogens in patients
- c) A gene that makes plants pest-resistant
- d) A herbicidal substance synthesized in genetically modified crops

Answer: A

- Cas9 (CRISPR associated protein 9) is a protein which plays a vital role in the immunological defense of certain bacteria against DNA viruses, and which is heavily utilized in genetic engineering applications.
- Cas9 is a bacterial RNA-guided endonuclease that uses base pairing to recognize and cleave target DNAs with complementarity to the guide RNA.
- The programmable sequence specificity of Cas9 has been harnessed for genome editing and gene expression control in many organisms.
- Its main function is to cut DNA and therefore it can alter a cell's genome

Question 5 (Previous Year Question 2019)

Q. Which one of the following statements is not correct?

- a) **Hepatitis B virus is transmitted much like HIV.**
- b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.
- c) Globally, the number of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.
- d) Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years.

Answer: **A**

- The hepatitis B vaccine is a safe and effective vaccine that is recommended for all infants at birth and for children up to 18 years.
- The hepatitis B vaccine is also recommended for adults living with diabetes and those at high risk for infection due to their jobs, lifestyle, living situations, or country of birth.
- Since everyone is at some risk, all adults should seriously consider getting the hepatitis B vaccine for a lifetime protection against a preventable chronic liver disease.
- The hepatitis B vaccine is also known as the first “anti-cancer” vaccine because it prevents hepatitis B, the leading cause of liver cancer worldwide. Thus option b is the answer.

Question 6 (Previous Year Question 2020)

Q. Consider the following statements:

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.
2. A person's genome can be edited before birth at the early embryonic stage.
3. Human /induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 2 only
- d) 1, 2 and 3**

Answer: D

- Germline gene therapy is the replacement of genes in egg or sperm cells with which an offspring inherits a new trait. It allows for the correction of disease- causing gene variants that are certain to be passed down from generation to generation. Hence,
- statement 1 is correct.
- CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) technology is employed to modify human embryos before they are transferred into women's uterus.
- Recently, researchers had successfully created the world's first genetically- edited baby. By using CRISPR technology, the genomes of embryos were edited to disable a gene, CCR5, which allows HIV to infect cells. Hence, statement 2 is correct.
- Due to certain anatomical and physiological features shared with humans, the pig is considered an important animal model of human diseases with unique advantages in surgery and xenotransplantation studies. Hence, statement 3 is correct.
- Therefore, option (d) is the correct answer.

Question 7 (Previous Year Question 2021)

Q. The term 'ACE2' is talked about in the context of

- a) genes introduced in the genetically modified plants
- b) development of India's own satellite navigation system
- c) radio collars for wildlife tracking
- d) spread of viral diseases**

Answer: D

- 'ACE2' is a proteinaceous enzyme on the surface of many cell types that stands for Angiotensin Convertase Enzyme-2. It is an enzyme that generates small proteins by cutting up the larger protein angiotensinogen that then goes on to regulate functions in the cell.
- Using the spike-like protein on its surface, the SARS-CoV-2 virus binds to ACE2 like a key being inserted into a lock – prior to entry and infection of cells. Hence, ACE2 acts as a cellular doorway or a receptor for the virus that causes COVID-19.
- Therefore, option (d) is the correct answer.

Question 8 (Previous Year Question 2022)

Q. Consider the following statements:

1. Biofilms can form on medical implants within human tissues.
2. Biofilms can form on food and food processing surfaces.
3. Biofilms can exhibit antibiotic resistance.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3**

Answer: D

- Biofilms are a collective of one or more types of microorganisms that can grow on many different surfaces.
- Microorganisms that form biofilms include bacteria, fungi and protists.
- One common example of a biofilm dental plaque, a slimy build-up of bacteria that forms on the surfaces of teeth.
- They can grow on plant tissues and animal tissues, on food and food processing surfaces and on implanted medical devices such as catheters and pacemakers.
- For microorganisms, living as a part of a biofilm comes with certain advantages. They are seen to exhibit antibiotic resistance.

Question 9 (Previous Year Question 2022)

Q. Consider the following statements in respect of probiotics :

1. Probiotics are made of both bacteria and yeast.
2. The organisms in probiotics are found in foods we ingest but they do not naturally occur in our gut.
3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) 1 and 3**
- (d) 2 and 3

Answer: C

- Probiotics are a combination of live beneficial bacteria and/or yeasts that naturally live in your body.
- Probiotics, live bacteria in our gut, can also be bought as supplements. They're renowned for their ability to boost nutrient absorption, fight inflammation, and enhance digestive health
- Lactobacillus genus of bacteria produces lactase, the enzyme that breaks down lactose, or milk sugar

Question 10 (Previous Year Question 2013)

Q. Which of the following diseases can be transmitted from one person to another through tattooing?

1. Chikungunya
2. Hepatitis B
3. HIV-AIDS

Select the correct answer using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only**
- d) 1, 2 and 3

Answer: C

- Hepatitis B is an infection of the liver. It can cause scarring of the organ, liver failure, and cancer.
- It can be fatal if it isn't treated.
- It's spread when people come in contact with the blood, open sores, or body fluids of someone who has the hepatitis B virus.
- HIV stands for human immunodeficiency virus.
- It harms your immune system by destroying the white blood cells that fight infection.
- This puts you at risk for serious infections and certain cancers. AIDS stands for Acquired Immunodeficiency Syndrome.
- It is the final stage of infection with HIV. Not everyone with HIV develops AIDS.
- HIV most often spreads through unprotected sex with an infected person.
- It may also spread by sharing drug needles or through contact with the blood of an infected person. Women can give it to their babies during pregnancy or childbirth.

Question 11 (Previous Year Question 2014)

Q. Consider the following diseases

1. Diphtheria
2. Chickenpox
3. Smallpox

Which of the above diseases has/have been eradicated in India?

- a) 1 and 2 only
- b) 3 only**
- c) 1, 2 and 3
- d) None

Answer: **B**

- Government introduced Pentavalent vaccine which also fights Diphtheria. It means Diphtheria is also prevalent.
- Chickenpox is still prevalent
- Smallpox has been eradicated from India

Question 12 (Previous Year Question 2014)

Q. Consider the following pairs :

Vitamin Deficiency disease

1. Vitamin C Scurvy
2. Vitamin D Rickets
3. Vitamin E Night blindness

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3
- d) None

Answer: **A**

- | | |
|--------------|---|
| 1. Vitamin C | Scurvy is right. |
| 2. Vitamin D | Rickets is right |
| 3. Vitamin E | Night blindness is wrong. Vitamin A causes Night blindness. |

Question 13 (Previous Year Question 2015)

Q. H1N1 virus is sometimes mentioned in the news with reference to which one of the following diseases?

- a) AIDS
- b) Bird flu
- c) Dengue
- d) Swine flu**

Answer: D

- Influenza (H1N1) virus is a subtype of influenza A virus that was the most common cause of human influenza (flu) in 2009 and is associated with the 1918 outbreak known as the Spanish Flu.
- It is an orthomyxovirus that contains the glycoproteins haemagglutinin and neuraminidase. For this reason, they are described as H1N1, H1N2 etc. depending on the type of H or N antigens they express with metabolic synergy.
- Haemagglutinin causes red blood cells to clump together and binds the virus to the infected cell.
- Neuraminidase is a type of glycoside hydrolase enzyme which helps to move the virus particles through the infected cell and assist in budding from the host cells
- In June 2009, the World Health Organization (WHO) declared the new strain of swine-origin H1N1 as a pandemic.